

Creation Date 16-Nov-2010

Revision Date 06-Oct-2023

Revision Number 7

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                                              |
|---------------------------|--------------------------------------------------------------|
| Product Description:      | <b>Mineral oil</b>                                           |
| Cat No. :                 | <b>415080000; 415080010; 415080025; 415080060; 415080100</b> |
| Synonyms                  | Paraffin oil                                                 |
| CAS No                    | 8042-47-5                                                    |
| EC No                     | 232-455-8                                                    |
| REACH registration number | 01-2119487078-27                                             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

#### EU entity/business name

Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

#### UK entity/business name

Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

#### Swiss distributor - Fisher Scientific AG

Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:

Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

Based on available data, the classification criteria are not met

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements

None required

### 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                    | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|------------------------------|-----------|-------------------|----------|---------------------------------------------------|
| White mineral oil, petroleum | 8042-47-5 | EEC No. 232-455-8 | 100      | -                                                 |

REACH registration number

01-2119487078-27

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

|                                           |                                                                                                                          |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.          |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.  |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                           |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                         |

## **4.2. Most important symptoms and effects, both acute and delayed**

No information available.

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

None under normal use conditions.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing.

### **6.2. Environmental precautions**

Avoid release to the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### **6.4. Reference to other sections**

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 10

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                    | Italy | Germany                                                                                                                                       | Portugal | The Netherlands | Finland |
|------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|---------|
| White mineral oil, petroleum |       | TWA: 5 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4<br>TWA: 5 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 mg/m <sup>3</sup> |          |                 |         |

| Component                    | Austria | Denmark | Switzerland                        | Poland | Norway |
|------------------------------|---------|---------|------------------------------------|--------|--------|
| White mineral oil, petroleum |         |         | TWA: 5 mg/m <sup>3</sup> 8 Stunden |        |        |

| Component                    | Estonia | Gibraltar | Greece | Hungary                               | Iceland |
|------------------------------|---------|-----------|--------|---------------------------------------|---------|
| White mineral oil, petroleum |         |           |        | TWA: 5 mg/m <sup>3</sup> 8 órában. AK |         |

| Component          | Latvia                   | Lithuania | Luxembourg | Malta | Romania |
|--------------------|--------------------------|-----------|------------|-------|---------|
| White mineral oil, | TWA: 5 mg/m <sup>3</sup> |           |            |       |         |

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

| petroleum                    |                                           |                 |                                                                                                                           |        |        |
|------------------------------|-------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Component                    | Russia                                    | Slovak Republic | Slovenia                                                                                                                  | Sweden | Turkey |
| White mineral oil, petroleum | Skin notation<br>MAC: 5 mg/m <sup>3</sup> |                 | TWA: 5 mg/m <sup>3</sup> 8 urah<br>respirable fraction<br>STEL: 20 mg/m <sup>3</sup> 15<br>minutah respirable<br>fraction |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                         | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|---------------------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| White mineral oil, petroleum<br>8042-47-5 ( 100 ) |                                 |                                    |                                   | DNEL = 217.05mg/kg<br>bw/day         |

| Component                                         | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|---------------------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| White mineral oil, petroleum<br>8042-47-5 ( 100 ) |                                     |                                        |                                       | DNEL = 164.56mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

No information available.

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|----------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability,

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particle filter

## Small scale/Laboratory use

Maintain adequate ventilation

**Recommended half mask:-** Particle filtering: EN149:2001

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                          |
|-----------------------------------------|--------------------------|------------------------------------------|
| Physical State                          | Liquid                   |                                          |
| Appearance                              | Clear, Viscous           |                                          |
| Odor                                    | No information available |                                          |
| Odor Threshold                          | No data available        |                                          |
| Melting Point/Range                     | No data available        |                                          |
| Softening Point                         | No data available        |                                          |
| Boiling Point/Range                     | No information available |                                          |
| Flammability (liquid)                   | No data available        |                                          |
| Flammability (solid,gas)                | Not applicable           | Liquid                                   |
| Explosion Limits                        | No data available        |                                          |
| Flash Point                             | > 220 °C / > 428 °F      | <b>Method -</b> No information available |
| Autoignition Temperature                | No data available        |                                          |
| Decomposition Temperature               | No data available        |                                          |
| pH                                      | No information available |                                          |
| Viscosity                               | 65-75 cSt at 40 °C       |                                          |
| Water Solubility                        | Insoluble                |                                          |
| Solubility in other solvents            | No information available |                                          |
| Partition Coefficient (n-octanol/water) |                          |                                          |
| Component                               | <b>log Pow</b>           |                                          |
| White mineral oil, petroleum            | 6                        |                                          |
| Vapor Pressure                          | No data available        |                                          |
| Density / Specific Gravity              | 0.877                    |                                          |
| Bulk Density                            | Not applicable           | Liquid                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                              |
| Particle characteristics                | Not applicable (liquid)  |                                          |

### 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

## 10.2. Chemical stability

Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

Hazardous polymerization does not occur.  
No information available.

## 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

## 10.5. Incompatible materials

None known.

## 10.6. Hazardous decomposition products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component                    | LD50 Oral         | LD50 Dermal          | LC50 Inhalation |
|------------------------------|-------------------|----------------------|-----------------|
| White mineral oil, petroleum | >5000 mg/kg (Rat) | >3000 mg/kg (Rabbit) | -               |

#### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

#### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

#### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

#### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

#### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

#### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

| Component                    | Freshwater Fish                                  | Water Flea | Freshwater Algae |
|------------------------------|--------------------------------------------------|------------|------------------|
| White mineral oil, petroleum | LC50: > 10000 mg/L, 96h<br>(Lepomis macrochirus) |            |                  |

### 12.2. Persistence and degradability

Persistence May persist.

### 12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate

| Component                    | log Pow | Bioconcentration factor (BCF) |
|------------------------------|---------|-------------------------------|
| White mineral oil, petroleum | 6       | No data available             |

### 12.4. Mobility in soil

The product is insoluble and floats on water. Spillage unlikely to penetrate soil. Is not likely mobile in the environment due to its low water solubility. Is not likely mobile in the environment due to its low water solubility and propensity to bind to soil particles.

### 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors.

### 12.7. Other adverse effects

Persistent Organic Pollutant  
Ozone Depletion Potential

This product does not contain any known or suspected substance.  
This product does not contain any known or suspected substance.

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

Waste from Residues/Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to



# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

ensure complete and accurate classification.

## Contaminated Packaging

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

## European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

## Other Information

Waste codes should be assigned by the user based on the application for which the product was used.

## Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required.

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                    | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| White mineral oil, petroleum | 8042-47-5 | 232-455-8 | -      | -   | X     | X    | KE-35412 | X    | X    |

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

| Component                    | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| White mineral oil, petroleum | 8042-47-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Authorisation/Restrictions according to EU REACH**

Not applicable

| Component                    | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| White mineral oil, petroleum | 8042-47-5 | -                                                                   | -                                                                             | -                                                                                                     |

**Seveso III Directive (2012/18/EC)**

| Component                    | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| White mineral oil, petroleum | 8042-47-5 | Not applicable                                                                            | Not applicable                                                                           |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**

Not applicable

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

**National Regulations**

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification**

See table for values

| Component                    | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|------------------------------|---------------------------------------|-------------------------|
| White mineral oil, petroleum | WGK1                                  |                         |

| Component                    | France - INRS (Tables of occupational diseases)         |
|------------------------------|---------------------------------------------------------|
| White mineral oil, petroleum | Tableaux des maladies professionnelles (TMP) - RG 36bis |

**Swiss Regulations**

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

# SAFETY DATA SHEET

Mineral oil

Revision Date 06-Oct-2023

## Full text of H-Statements referred to under sections 2 and 3

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Creation Date** 16-Nov-2010  
**Revision Date** 06-Oct-2023  
**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**